Re-analysis Summary Report for Peer-evaluation

Paper ID: **McDevitt_JournPoliEco_2014_yQeR**

Re-analyst's ID: **xsxir**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
The average plumbing firm whose name begins with A or a number receives… more service complaints than other firms …(p. 909.)

The corresponding article

Here you can find the original article: https://osf.io/6fusy/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/z7mfu/?view_only=277b7d18bb24436eb5cfca69a07eb07a

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
In order to test the main hypothesis, whether plumbing firms with names starting with A or number receive more complaints than rest of firms, I first analyzed causal diagram for possible confounds. Although majority of variables (firm size, number of employees, whether the company is on Google) does not need to be controlled for, number of different names that company uses is one thing that needs to be incorporated into analyses. Original paper stated that number of complaints was summed for the same company using different name and if only one of the names started as A, this company was also treated as having an A in its name. Therefore, it is reasonable to assume that companies could use this as strategy and this variable should be used in the model as well. For the analysis, we used linear model with two independent variables. First (nominal) codes whether name of plumbing company starts with A or number, or it does not. Second continuous variable corresponds to number of names that the company used. Number of complaints was used as independent variable. Both variables were z-transformed before the analysis to obtain standardized beta coefficients.

The analyst's conclusion: The number of complaints did not differ between companies with name starting with A or letter and between the rest (beta = -0.01, p = 0.798). The difference in number of complaints can be expressed with number of names that the company used (beta = 0.49, $p < 0.001$). Thus, although direct comparison of number of complaints would reveal differences between companies (similarly as found in the original study), this difference is completely explained by number of names. This could reflect a strategy by the companies to have at least one company with A name, so customers would find them first in the yellow pages.

The analyst's categorization of the result:
The results do not show evidence for or against the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
Use only the data of Illinois Plumbing Firms in your model. Do not restrict your sample to those firms that serve the metro Chicago area. Do not include the reviews received via yelp.com, Angie's List, and Consumer's Checkbook into your analysis. You should use the overall complaints number instead of complaints per employee.

Here, the analyst reported the following steps and results of the analysis:
In order to test the main hypothesis, whether plumbing firms with names starting with A or number receive more complaints than rest of firms, I first analyzed causal diagram for possible confounds. Although majority of variables (firm size, number of employees, whether the company is on Google) does not need to be controlled for, number of different names that company uses is one thing that needs to be incorporated into analyses. Original paper stated that number of complaints was summed for the same company using different name and if only one of the names started as A, this company was also treated as having an A in its name. Therefore, it is reasonable to assume that companies could use this as strategy and this variable should be used in the model as well. For the analysis, we used linear model with two independent variables. First (nominal) codes whether name of plumbing company starts with A or number, or it does not. Second continuous variable corresponds to number of names that the company used. Number of complaints was used as independent variable. Both variables were z-transformed before the analysis to obtain standardized beta coefficients. For final reporting, we used F-test to test the significance of the variable coding whether name of plumbing company starts with A or number, or it does not.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/5pgqv/